Mathematical Notation Guide for peppwR Methods

Explanatory Guide

March 25, 2026

Introduction

This guide explains all mathematical formulas and notation used in the Methods section of the peppwR JSS article. Each formula is presented with plain-language explanations of what the symbols mean, how to read the notation, what the formula does conceptually, and why it's used in the peppwR context.

1 Statistical Framework Section

1.1 Formula 1: Statistical Power Definition

Mathematical notation:

$$P(\text{reject } H_0 | H_1 \text{ true}) = 1 - \beta$$

What the symbols mean:

- $P(\cdot)$ = probability of something happening
- H_0 = null hypothesis (no difference between groups)
- H_1 = alternative hypothesis (there is a difference between groups)
- β = Type II error rate (probability of missing a real effect)
- $|$ = "given that" or "conditional on"

How to read it: "Power equals the probability of rejecting the null hypothesis, given that the alternative hypothesis is true, which equals 1 minus beta"

What it does: Defines statistical power as the chance of correctly detecting a real effect when one actually exists

Why peppwR uses it: This is the fundamental concept that peppwR calculates through simulation - the probability that your experiment will successfully detect a real biological effect

1.2 Formula 2: Hypothesis Testing Framework

Mathematical notation:

$$H_0 : \mu_{\text{treatment}} = \mu_{\text{control}} \quad \text{vs} \quad H_1 : \mu_{\text{treatment}} \neq \mu_{\text{control}}$$

What the symbols mean:

- $\mu_{\text{treatment}}$ = mean abundance in the treatment group
- μ_{control} = mean abundance in the control group
- $=$ = equals (no difference)
- $\neq$ = not equals (there is a difference)

How to read it: "Null hypothesis: treatment mean equals control mean, versus alternative hypothesis: treatment mean does not equal control mean"

What it does: Sets up the statistical question we're trying to answer - is there a difference between groups?

Why peppwR uses it: This is the standard framework for comparing peptide abundances between experimental conditions

1.3 Formula 3: Sample Size Determination

Mathematical notation:

$$n^* = \min\{n : P(\text{reject } H_0 | \delta, n) \geq \pi_{\text{target}}\}$$

What the symbols mean:

- n^* = optimal sample size
- $\min\{\cdot\}$ = minimum value that satisfies the condition
- n = sample size (number of samples per group)
- δ = effect size (how big the difference is)
- π_{target} = target power level (usually 0.8 for 80% power)
- $\geq$ = greater than or equal to

How to read it: "n-star equals the minimum n such that the probability of rejecting the null hypothesis, given delta and n, is greater than or equal to the target power"

What it does: Finds the smallest sample size that will give you your desired statistical power

Why peppwR uses it: This answers the key question "How many samples do I need?" for your experiment

1.4 Formula 4: Power Estimation

Mathematical notation:

$$\pi(n, \delta) = P(\text{reject } H_0 | \delta, n)$$

What the symbols mean:

- $\pi(n, \delta)$ = power as a function of sample size and effect size
- $P(\cdot)$ = probability
- The rest as defined above

How to read it: "Pi of n and delta equals the probability of rejecting the null hypothesis given delta and n"

What it does: Calculates the statistical power for a given sample size and effect size

Why peppwR uses it: This answers "What power will my planned experiment have?"

1.5 Formula 5: Minimum Detectable Effect Size

Mathematical notation:

$$\delta^* = \min\{\delta : P(\text{reject } H_0 | \delta, n) \geq \pi_{\text{target}}\}$$

What the symbols mean:

- δ^* = minimum detectable effect size
- Other symbols as defined previously

How to read it: "Delta-star equals the minimum delta such that the probability of rejecting the null hypothesis, given delta and n, is greater than or equal to target power"

What it does: Finds the smallest biological effect you can reliably detect with your sample size

Why peppwR uses it: This answers "What's the smallest effect I can detect?" given your experimental constraints

1.6 Formula 6: Peptidome-wide Power

Mathematical notation:

$$P_{\text{peptidome}}(n, \delta, \pi_{\text{target}}) = \frac{1}{K} \sum_{i=1}^{K} \mathbf{1}[\pi_i(n, \delta) \geq \pi_{\text{target}}]$$

What the symbols mean:

- $P_{\text{peptidome}}$ = proportion of peptides achieving target power

- K = total number of peptides
- $\sum_{i=1}^{K}$ = sum from peptide 1 to peptide K
- $\mathbf{1}[\cdot]$ = indicator function (equals 1 if condition is true, 0 if false)
- $\pi_i(n, \delta)$ = power for peptide i

How to read it: "Peptidome power equals one over K times the sum of indicator functions for whether each peptide's power exceeds the target"

What it does: Calculates the fraction of peptides that will achieve your target power level

Why peppwR uses it: Since different peptides have different properties, this tells you what percentage of your peptides will be detectable

1.7 Formula 7: FDR Control (Benjamini-Hochberg)

Mathematical notation:

$$k^* = \max\left\{i : p_{(i)} \leq \frac{i \cdot \alpha}{m}\right\}$$

What the symbols mean:

- k^* = cutoff index for rejecting hypotheses
- $\max\{\cdot\}$ = maximum value satisfying the condition
- $p_{(i)}$ = i-th smallest p-value (ordered from smallest to largest)
- α = significance level (usually 0.05)
- m = total number of statistical tests performed

How to read it: "k-star equals the maximum i such that the i-th ordered p-value is less than or equal to i times alpha divided by m"

What it does: Determines which p-values to call significant when testing many peptides simultaneously

Why peppwR uses it: When testing thousands of peptides, you need to adjust for multiple comparisons to avoid false discoveries

2 Distribution Fitting Section

2.1 Formula 8: Maximum Likelihood Estimation

Mathematical notation:

$$\hat{\theta}_{k,i} = \arg\max_{\theta_k} \sum_{j=1}^{n_i} \log f_k(x_{ij}|\theta_k)$$

What the symbols mean:

- $\hat{\theta}_{k,i}$ = estimated parameters for distribution k and peptide i
- $\arg\max$ = "the argument that maximizes" - find the parameter values that make the expression biggest
- θ_k = parameters of distribution k
- $\sum_{j=1}^{n_i}$ = sum over all j from 1 to n_i (all observations for peptide i)
- log = natural logarithm
- $f_k(x_{ij}|\theta_k)$ = probability density function for distribution k
- x_{ij} = j-th observation for peptide i

How to read it: "Theta-hat k,i equals the argument that maximizes the sum of log probability densities"

What it does: Finds the distribution parameters that make the observed data most likely

Why peppwR uses it: This is how we fit mathematical distributions (like gamma or lognormal) to each peptide's abundance data

2.2 Formula 9: Model Selection (AIC)

Mathematical notation:

$$\hat{k}_i = \arg\min_k \left\{ -2 \sum_{j=1}^{n_i} \log f_k(x_{ij}|\hat{\theta}_{k,i}) + 2p_k \right\}$$

What the symbols mean:

- $\hat{k}_i$ = best distribution choice for peptide i
- $\arg\min$ = "the argument that minimizes" - find the k that makes the expression smallest
- -2 = constant multiplier in AIC formula
- $2p_k$ = penalty term where p_k is the number of parameters in distribution k

How to read it: "k-hat i equals the argument that minimizes negative two times the log-likelihood plus two times the number of parameters"

What it does: Chooses the best distribution for each peptide, balancing goodness of fit against complexity

Why peppwR uses it: AIC helps select whether gamma, lognormal, normal, or inverse Gaussian best describes each peptide's data

3 Simulation Engine Section

3.1 Formula 10: Monte Carlo Power Estimation

Mathematical notation:

$$\hat{\pi}_i(n, \delta) = \frac{1}{n_{\text{sim}}} \sum_{j=1}^{n_{\text{sim}}} R^{(j)}$$

What the symbols mean:

- $\hat{\pi}_i(n, \delta)$ = estimated power for peptide i
- n_{sim} = number of simulations (usually 1000)
- $R^{(j)}$ = result of simulation j (1 if significant, 0 if not)
- $\sum_{j=1}^{n_{\text{sim}}}$ = sum over all simulations

How to read it: "Pi-hat i equals one over n-sim times the sum of all simulation results"

What it does: Estimates power by running many simulated experiments and counting how often they detect the effect

Why peppwR uses it: Since analytical power calculations are impossible for complex proteomics data, simulation provides accurate estimates

3.2 Formula 11: Monte Carlo Standard Error

Mathematical notation:

$$\text{SE}(\hat{\pi}_i) = \sqrt{\frac{\hat{\pi}_i(1 - \hat{\pi}_i)}{n_{\text{sim}}}}$$

What the symbols mean:

- $\text{SE}(\hat{\pi}_i)$ = standard error of the power estimate
- $\sqrt{\cdot}$ = square root
- $(1 - \hat{\pi}_i)$ = one minus the power estimate

How to read it: "Standard error of pi-hat equals the square root of pi-hat times one minus pi-hat, divided by the number of simulations"

What it does: Calculates the uncertainty in our power estimate due to simulation randomness

Why peppwR uses it: This tells you how precise your power estimate is - more simulations give lower standard error

4 Missing Data Handling Section

4.1 Formula 12: Mean Abundance Calculation

Mathematical notation:

$$\bar{x}_i = \frac{1}{n_{\text{obs},i}} \sum_{j:x_{ij} \neq \text{NA}} x_{ij}$$

What the symbols mean:

- $\bar{x}_i$ = mean abundance for peptide i
- $n_{\text{obs},i}$ = number of observed (non-missing) values for peptide i
- $\sum_{j:x_{ij} \neq \text{NA}}$ = sum over only the non-missing observations
- NA = missing value (Not Available)

How to read it: "x-bar i equals one over n-observed,i times the sum of all non-missing x values"

What it does: Calculates the average abundance for a peptide, excluding missing values

Why peppwR uses it: Missing data is common in proteomics, so we calculate means from available data only

4.2 Formula 13: Missingness Rate

Mathematical notation:

$$r_i = \frac{n_{\text{missing},i}}{n_{\text{total},i}}$$

What the symbols mean:

- r_i = missingness rate for peptide i
- $n_{\text{missing},i}$ = number of missing values for peptide i
- $n_{\text{total},i}$ = total number of samples for peptide i

How to read it: "r i equals the number of missing values divided by the total number of values"

What it does: Calculates what fraction of data is missing for each peptide

Why peppwR uses it: Different peptides have different amounts of missing data, which affects power calculations

4.3 Formula 14: Missing Data in Simulations

Mathematical notation:

$$\lfloor n \cdot r_i \rfloor$$

What the symbols mean:

- $\lfloor \cdot \rfloor$ = floor function (rounds down to nearest integer)
- n = sample size
- r_i = missingness rate for peptide i

How to read it: "Floor of n times r i"

What it does: Calculates how many observations to randomly remove from simulated data

Why peppwR uses it: This recreates realistic missing data patterns in power simulations

5 Summary

This guide covers all major mathematical notation in the peppwR Methods section. The formulas fall into four main categories:

1. **Statistical Framework:** Defines power, hypothesis tests, and optimization problems
2. **Distribution Fitting:** Uses maximum likelihood and AIC for model selection
3. **Simulation Engine:** Estimates power through Monte Carlo simulation
4. **Missing Data:** Characterizes and incorporates missingness patterns

The mathematical notation enables precise description of peppwR's algorithms while the simulations handle the complexity that makes analytical solutions impossible for proteomics data.